Jules Van Cauwenberghe

MSc Mechanical Engineering, ETH Zurich | Energy Systems, Risk & Optimization
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Seeking a Semester Project in energy-system modelling, optimization & risk

Profile

MSc student in Mechanical Engineering at ETH Zurich (Energy track), with a BSc from EPFL. Applies system-level risk analysis to railway safety projects under European and Swiss regulations and builds quantitative models (FEA, Python, MATLAB). Currently leading a team of 5 on a project for CIMO to decarbonize an industrial steam supply.

Education

ETH Zurich (ETHZ) — Zurich, Switzerland
MSc in Mechanical Engineering – Energy Specialization (Tutor: Prof. Russell McKenna) — **Sep 2026 – Present**

- **Autumn 2026:** Modeling & Optimization of Sustainable Energy Systems, Energy Systems Analysis, Stochastic Methods for Engineers, Power Market & Risk Management, Renewable Energy Technologies, Nuclear Energy Conversion, Information Systems for Engineers
- **Planned (spring 2027):** Advanced Techniques for Risk Analysis of Technical Systems, Optimization & ML

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
Bachelor of Science in Mechanical Engineering — **Sep 2023 – Jun 2026**

- **Relevant coursework:** Probability & Statistics, Numerical Analysis, Optimization & Operations Research, Control Systems, Foundations of AI, Thermodynamics, Finite Element Method
- **Academic project:** Built an autonomous fruit-harvesting gripper (5-person team), leading embedded control, sensing, UI and actuation (Arduino, OpenCV); ranked 4th out of 51 teams

Experience

Energy Now! (ETH Zurich) – ADAPT Project for CIMO — Zurich, Switzerland
Team Leader & Energy System Modeler (team of 5) — **Oct 2026 – Present (ongoing)**

- Assess heat pumps and e-boilers to decarbonize CIMO's 400 GWh/yr steam supply at Monthey, under a reliability constraint of at most 4 supply interruptions per year, each lasting up to 6 h
- Lead a team of 5 and the interface with CIMO stakeholders; formulate asset-sizing and dispatch optimization models (decision variables, constraints, cost objective) based on clustered typical-week profiles
- Develop a report and 2 models (portfolio sizing, real-time dispatch), in progress and due 14 Dec 2026, for a 2035 scenario with storage and a 130 kt/yr steam gap, on 6 typical weeks at 1-min resolution

Transports Lausannois (TL) — Lausanne, Switzerland
Industry-Sponsored Mechanical Engineering Internship – Railway Safety & Systems Risk — **Feb 2026 – May 2026**

- Identified a passenger-safety gap on the M1 metro line (standing area with no accessible handrail) and initiated the collaboration with TL; led a 2-person team through a system redesign for the 22-unit fleet
- Led system-level risk and interface analysis across 3 rolling-stock generations, ensuring compliance with 5 European (EN) standards (fire and smoke, structural loads, accessibility) and Swiss LHand/OFT regulations
- Developed suspended-handrail concepts in CAD and structural FEA (fatigue safety factors of 4.2–7.8 under normal loads); delivered a dossier preparing industrialization: formal risk analysis, cost estimate, 8 qualified Swiss suppliers

EPFL Xplore – European Rover Challenge (ERC) — Lausanne, Switzerland
Mechanical Design Engineer – Robotic Arm Hardware — **Sep 2025 – Jun 2026**

- Designed the structure of the robotic arm of EPFL Xplore's rover as sole structural engineer; the team won the European Rover Challenge (ERC) 2026 in Kraków, its second consecutive title
- Validated the structure by FEA under system-level load cases (70/35 Nm joint torques, 310 N); folded sheet-metal and welded architecture cut subsystem mass by 750 g (–25%) while maintaining stiffness

Technical Skills

Modelling & Analysis: **Risk analysis**, system-level modelling, **FEA**, numerical methods

Programming: **Python**, MATLAB, C, LabVIEW

Engineering Tools: CATIA, Fusion 360, Inventor, Abaqus

Industrial Context: Railway & energy standards (EN/ISO/OFT/LHand), regulatory compliance, cost estimation

Additional

Academic tutoring – mathematics & physics

Competitive rugby – highest national youth level in France (Crabos), Normandy regional selection

Languages

French Native
English Professional working proficiency
Spanish Intermediate proficiency